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# MODEL 53401 EPG SINGLE CHAMBER TEMPORARY EXTERNAL PACEMAKER

In-service Guide

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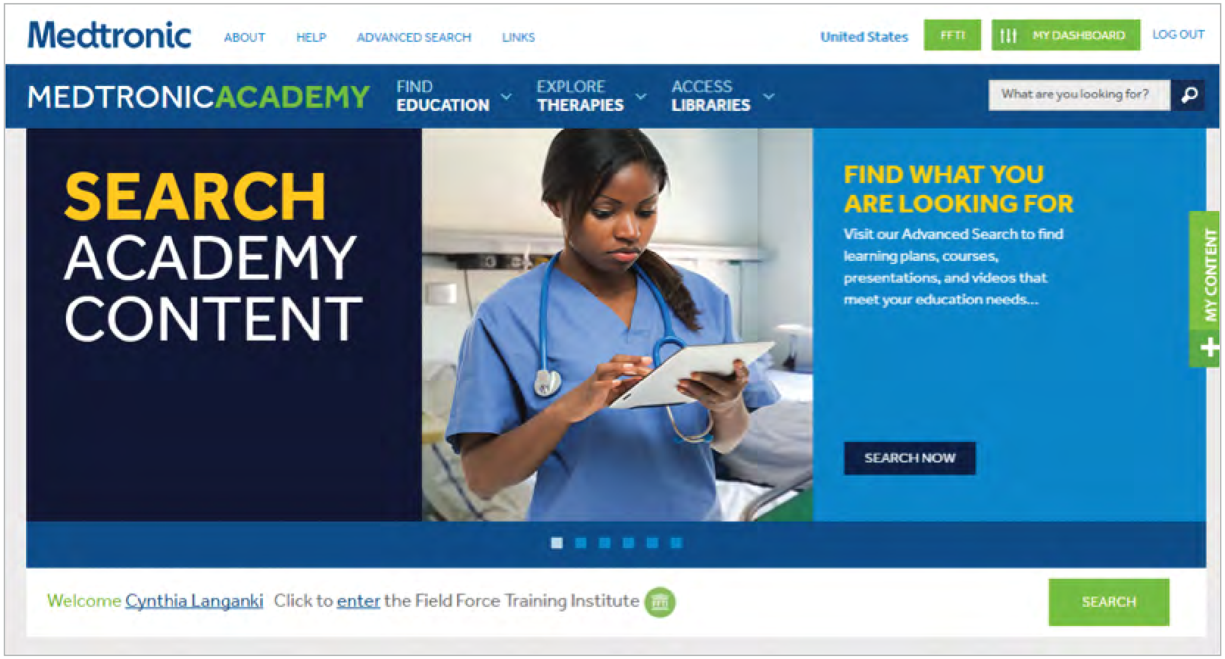
# 53401 TRAINING
## LEARNING OBJECTIVES

- Identify the control knobs and be able to explain how to use them to program device.
- Distinguish oversensing, undersensing, and loss of capture.
- Identify the basic components of the 53401 Temporary External Pacemaker.
- Identify the basic steps of Rapid Atrial Pacing.
- Locate resources for the 53401 EPG (e.g., Cable Compatibility Guide, other training materials).
- Describe the key steps in changing the batteries.
- Identify do's and don'ts for cleaning the device.
- Identify compatible cables.
- Describe when the device should be returned to Medtronic for cleaning or maintenance.

# 53401 TRAINING RESOURCES






**Education materials:**

medtronic academy.com

**Videos:**

medtronic academy.com

youtube.com

**Manuals:**

manuals. medtronic.com/ manuals/main/ region

## Intended Use

- Used with a cardiac pacing lead system for temporary single chamber pacing in a clinical environment by trained personnel
- Used where short-term demand (synchronous) or asynchronous pacing is indicated for therapeutic, prophylactic, or diagnostic purposes
- Must be used in an environment where the patient is monitored continuously to ensure that it is operating properly and delivering appropriate therapy to the patient

OVERVIEW

BASIC OPERATION

TROUBLESHOOTING PROCESS

RAPID ATRIAL PACING (RAP)

COMPONENTS: DEVICE, BATTERY, AND CABLE

MAINTENANCE AND CLEANING

SUMMARY

## Contraindications

There are no known contraindications for the use of temporary pacing as a means to control heart rate. However, the patient's age and medical condition may dictate the type of temporary pacemaker and lead system that the physician uses.

**Note:** Temporary pacing is contraindicated in the presence of another pacing system.

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OVERVIEW

BASIC OPERATION

TROUBLESHOOTING PROCESS

RAPID ATRIAL PACING (RAP)

COMPONENTS: DEVICE, BATTERY, AND CABLE

MAINTENANCE AND CLEANING

SUMMARY

# SYSTEM SCOPE
## SINGLE CHAMBER EPG SYSTEM

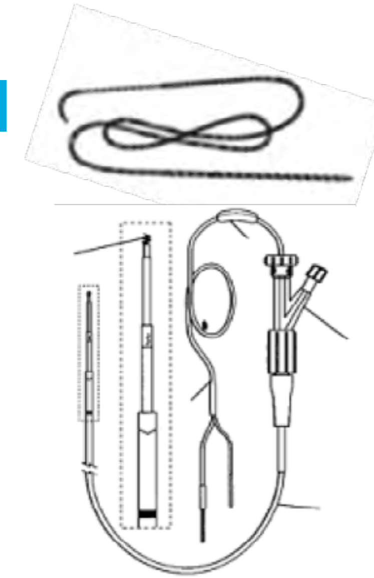
1


Model 53401
Single Chamber EPG

2

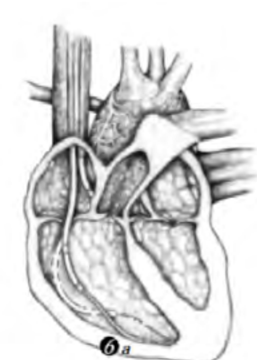
Patient Cable
(Model 5433)

3



Heart Wires/
Temporary Pacing Leads

4



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# BASIC OPERATION
## BUTTONS, DIALS, AND INDICATORS


**Buttons:**

1 On/Off

2 Lock/Unlock

**Parameter Dials:**

3 Rate

4 Output

5 Sensitivity

6 Rapid Atrial Pacing Controls

**Indicators:**

7 Battery Indicator

8 Lock Indicator

# BASIC OPERATION
## POWER BUTTON

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- Power button turns device On and Off.
- To turn Off, press and hold the Power button for 2 seconds.*
- If not held for 2 seconds, the temporary pacemaker will remain On.

*Device needs to be unlocked to turn Off.

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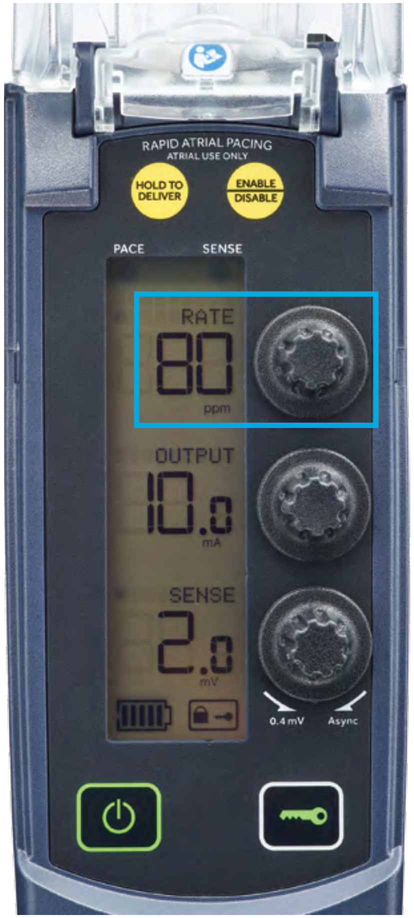
# BASIC FUNCTION
## LOCK BUTTON


- Press the lock button to hold settings.
- Lock indicator shows if controls are locked.
- Device needs to be Unlocked to reprogram and turn device Off.
- Device automatically locks **60 seconds** after the last parameter adjustment.

# BASIC FUNCTION
## PACING RATE CONTROL






- Select the desired pacing rate by turning the rate button, top knob.
- Turn clockwise for higher rates (counter clockwise for lower rates).
- The range is 30-200 paces per minute (ppm).

# BASIC FUNCTION
## VENTRICULAR OR ATRIAL PACING

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OVERVIEW

BASIC OPERATION

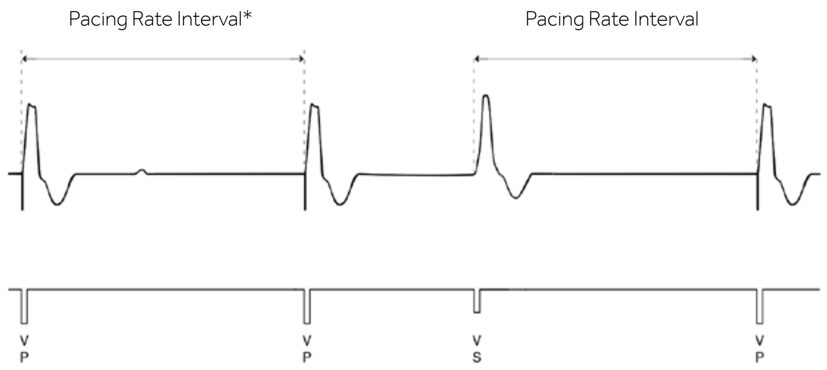
TROUBLESHOOTING PROCESS

RAPID ATRIAL PACING (RAP)

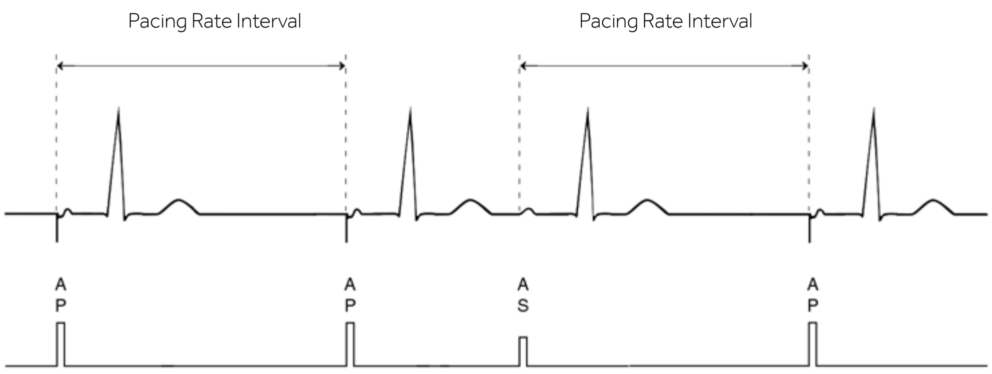
COMPONENTS: DEVICE, BATTERY, AND CABLE

MAINTENANCE AND CLEANING

SUMMARY


**Ventricular: VVI pacing**
Paces only the ventricles.




**Atrium: AAI pacing**
- Paces only the atrium.
- The stimulation conducts through the AV node to the ventricles.
- Not for use in patients with high degree heart block.

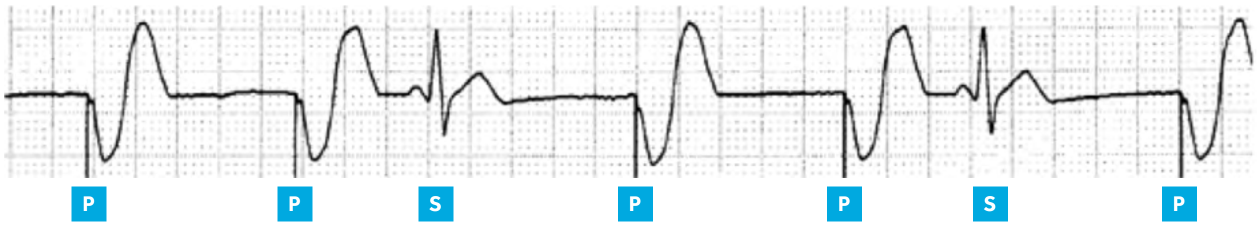
*The pacing rate interval may time from a paced or an intrinsic beat.

# BASIC FUNCTION
## SENSING AND PACING



### Sensing Patient's intrinsic rhythm: No Pacing

### Pacing and Sensing




P = Paced beat
S = Sensed (intrinsic beat)

# BASIC FUNCTION
## SENSING AND PACING

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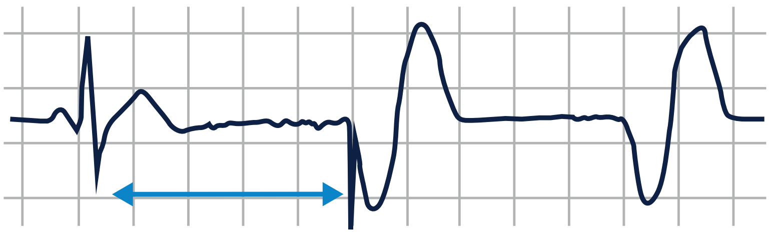
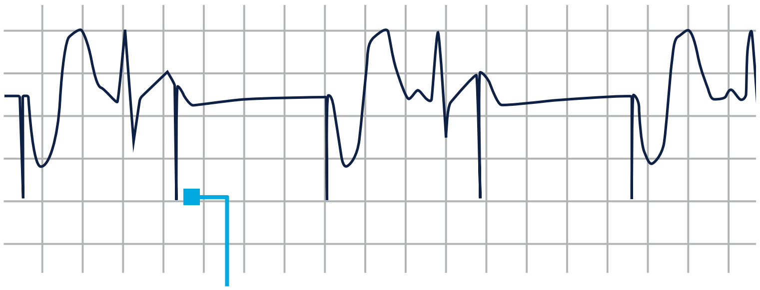


Pacing rate set from intrinsic beat.



Paced beat unchanged from intrinsic beat.

### Synchronous and Asynchronous Pacing

Inhibits pacing for intrinsic beats

### Asynchronous Pacing

Pacemaker will emit an output at a fixed rate regardless of intrinsic activity.

**Note:** Asynchronous pacing is contraindicated in the presence of intrinsic cardiac rhythms.

**Caution:** Because it may compete with the intrinsic activity of the heart, asynchronous pacing may result in tachyarrhythmia.

# BASIC FUNCTION
OUTPUT

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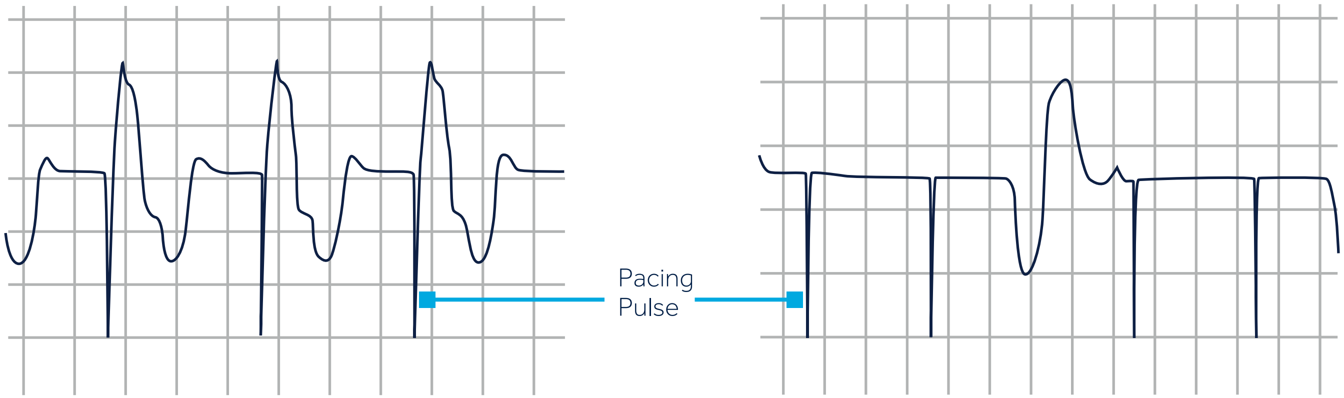
- Energy used to produce a heartbeat
- Constant current device
- Range: 0.1-25 milliamps (mA)

# BASIC FUNCTION
## STIMULATION CAPTURE






**Capture**
Depolarization of cardiac tissue after a pacing pulse. ECG shows a P wave or QRS after the pacing pulse.

**Loss of Capture**
ECG shows no heart response as a result of the pacing pulse.

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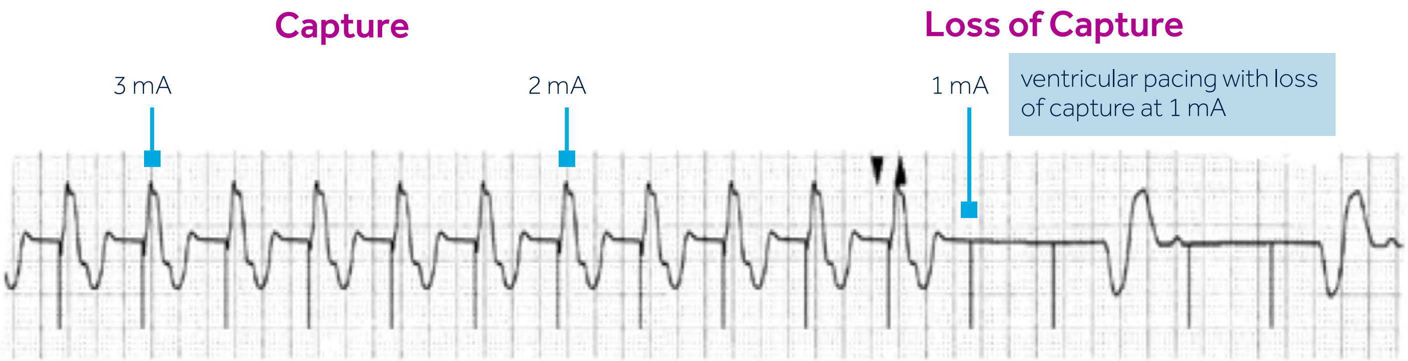
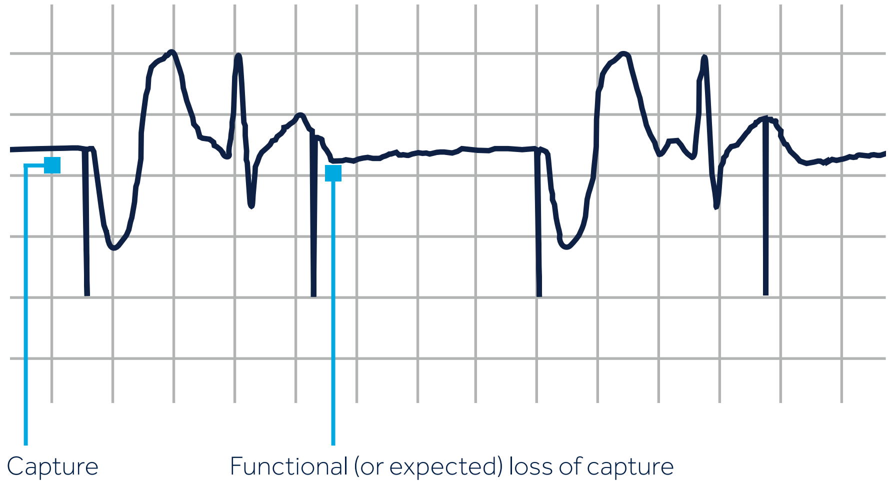
# BASIC PACING
## CAPTURE THRESHOLD




The minimum electrical stimulus needed to consistently capture the heart outside of the heart's own refractory period

# BASIC PACING
## FUNCTIONAL NON-CAPTURE






If the pacemaker paces too close to an intrinsic beat there may be expected non-capture because the cardiac tissue is in refractory from the intrinsic beat.

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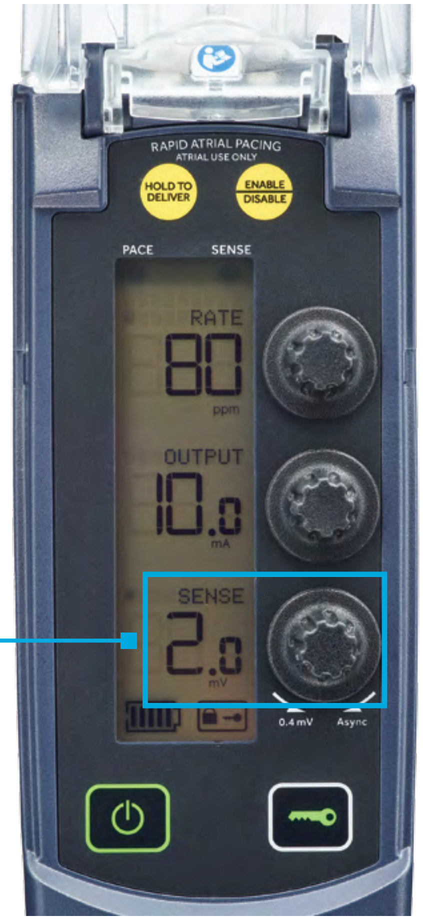
# BASIC FUNCTION
## SENSITIVITY CONTROL




Sensitivity Setting

- Higher number = less sensitive
- Lower number = more sensitive

- Used by the pacemaker to detect (or "see") the patient's intrinsic beats
- Needed for synchronous pacing
- Range: 0.4-20.0 mV

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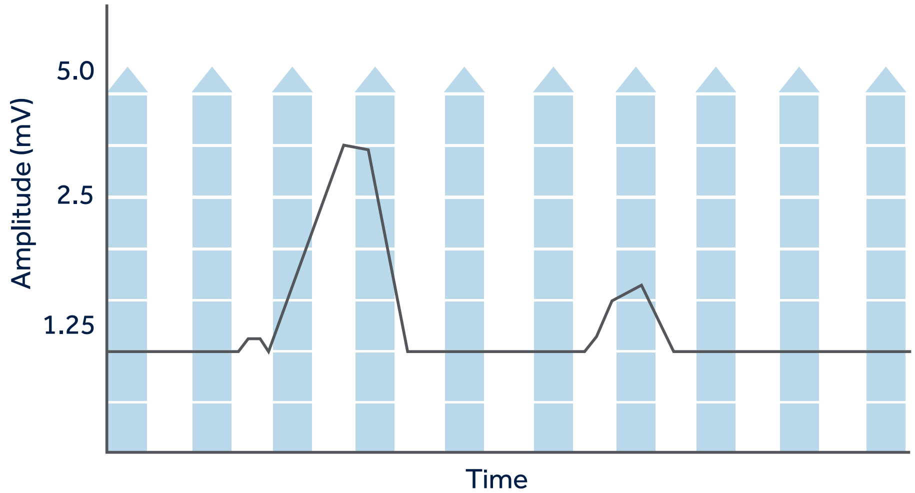
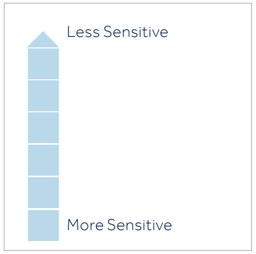
- OVERVIEW
- BASIC OPERATION
- TROUBLESHOOTING PROCESS
- RAPID ATRIAL PACING (RAP)
- COMPONENTS: DEVICE, BATTERY, AND CABLE
- MAINTENANCE AND CLEANING
- SUMMARY

# BASIC PACING FUNCTION
## SENSITIVITY OVERVIEW

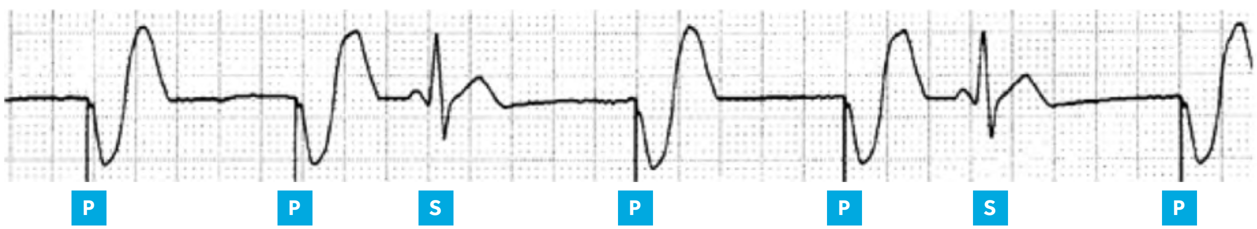



Sensing is what the pacemaker "sees." Like a fence, with a higher number less is sensed, with a lower number, more is sensed.

This R wave would not be sensed because the sensitivity is set at too high of a number.

### Pacing and Sensing




**P = Paced beat**
**S = Sensed (intrinsic beat)**

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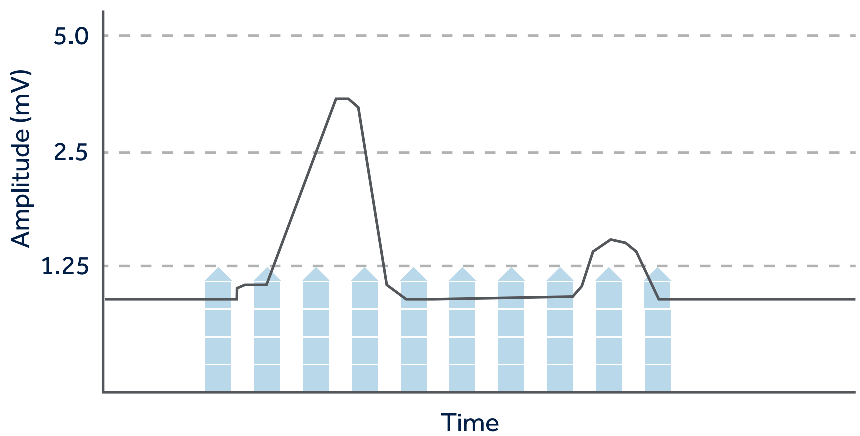
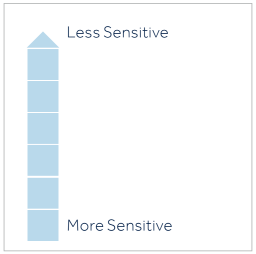
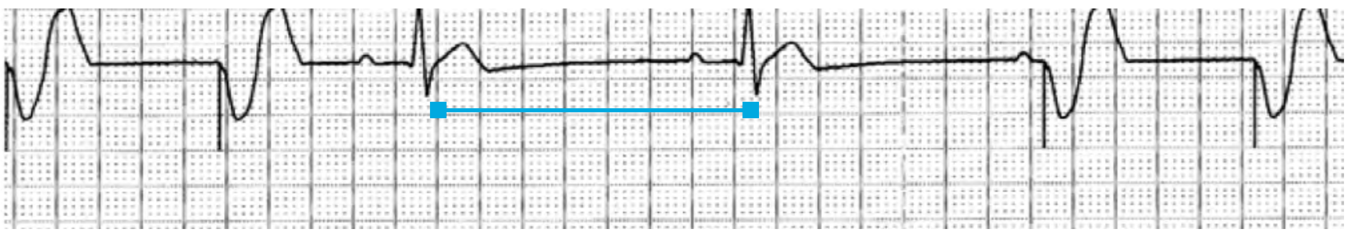
# BASIC PACING FUNCTION
## SENSITIVITY OVERVIEW




This R wave is sensed however so is the T wave as the sensitivity is set at too low of a number. This will cause oversensing.



Settings:
Mode: VVI Rate: 60 bpm

**Oversensing:** May occur if sensitivity set too sensitive. This results in under pacing. Note: On this ECG the paced beat did not occur at the expected time causing the rate to drop below the programmed rate.

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# THE REVISED NASPE/BPEG GENERIC CODE FOR (SINGLE CHAMBER) ANTIBRADYCARDIA PACING

## 1st Letter
**Chamber(s) Paced**

**A** = atrium
**V** = ventricle

## 2nd Letter
**Chamber(s) Sensed**

**A** = atrium
**V** = ventricle
**O** = none

## 3rd Letter
**Response to Sensing**

**I** = inhibit (Demand mode)
**T** = triggered
**O** = none (Asynch)

**V** **V** **I**

# BASIC INSTRUCTIONS
## PACING SETUP TABLE

| Pacing Setup Table | AOO | VOO | AAI | VVI | RAP* Rapid Atrial Pacing |
|---|---|---|---|---|---|
| **RATE Range** | 30-200 ppm | 30-200 ppm | 30-200 ppm | 30-200 ppm | 80-800 ppm |
| **SENSITIVITY** | **ASYNC** | **ASYNC** | 0.4-20 mV | 0.4-20 mV | NA Asynchronous during RAP |
| **OUTPUT** | 0.1-25 mA | 0.1-25 mA | 0.1-25 mA | 0.1-25 mA | 0.1-25 mA |
| **OTHER STEPS** | | | | | 1. Depress **"Enable"** to adjust settings<br>2. Press **"Hold to Deliver"** to start pacing |

*__Caution__: RAP is for atrial use only. Before enabling RAP, be sure that the atrial leads are connected to the atrium, not the ventricle.

# BASIC FUNCTION
## KNOWLEDGE CHECK



**Match the term on the left with the correct definition from the right**

____ Capture Threshold

____ Synchronous Pacing

____ Asynchronous Pacing

A. The minimum output pulse needed to consistently capture the heart

B. Pacemaker will emit an output at a fixed rate regardless of intrinsic activity

C. Inhibits pacing in the presence of intrinsic beats

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# BASIC FUNCTION
## KNOWLEDGE CHECK

**Match the term on the left with the correct definition from the right**

__A.__ Capture Threshold

__C.__ Synchronous Pacing

__B.__ Asynchronous Pacing

A. The minimum output pulse needed to consistently capture the heart

B. Pacemaker will emit an output at a fixed rate regardless of intrinsic activity

C. Inhibits pacing in the presence of intrinsic beats

# BASIC FUNCTION
## KNOWLEDGE CHECK

**Which of the following is correct for a device programmed VVI?**

A. The device will pace the ventricle, sense in the atrium, and inhibit in response to intrinsic beats.

B. The device will pace and sense the ventricle and inhibit in response to intrinsic beats.

C. The device will pace and sense the ventricle and have no response to intrinsic beats.

# KNOWLEDGE CHECK ANSWER

**Which of the following is correct for a device programmed VVI?**

A. The device will pace the ventricle, sense in the atrium, and inhibit in response to intrinsic beats.

**B. The device will pace and sense the ventricle and inhibit in response to intrinsic beats.**

C. The device will pace and sense the ventricle and have no response to intrinsic beats.

Feedback: The first letter refers to what is paced, the second letter refers to what is sensed, and the third letter is for the action. V stands for ventricular in the NASPE/BPEG code.
Review section: The Revised NASPE/BPEG Generic Code for (Single Chamber) Antibrady pacing.

# TROUBLESHOOTING PROCESS

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**1. Gather Information:**

A. ECG Strip

B. Pacing rate

C. Sensitivity setting

D. Output setting

E. Battery status

**2. Identify the Problem and Possible Cause:**

A. Check cable and lead connections.

**3. Identify the Solution and Carry Out Corrective Procedures**

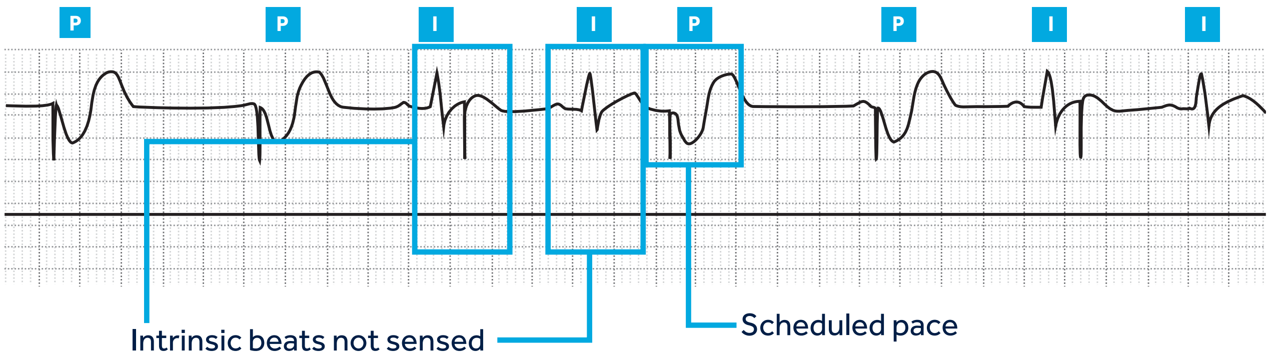
A. Follow specific steps if sensing or capture issue.

# TROUBLESHOOTING
## UNDERSENSING . . . OVER PACING

VVI/60

Pacemaker does not "see" the intrinsic beat, and therefore does not respond appropriately

### Pacing and Sensing

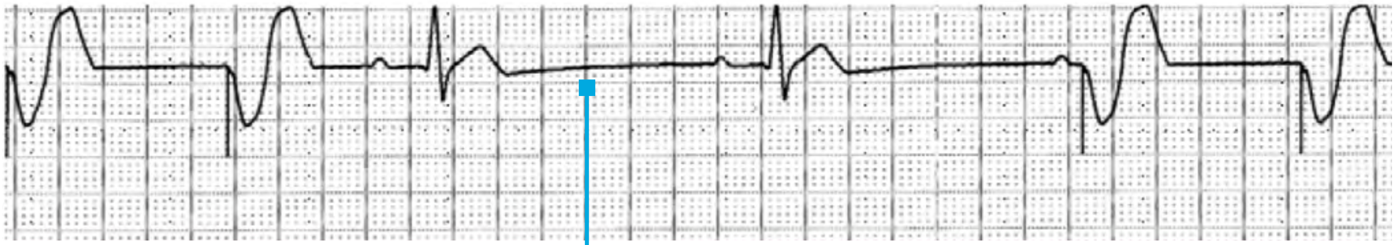



P = Paced beat
I = Intrinsic beat

# TROUBLESHOOTING
## OVERSENSING . . . UNDER PACING

VVI/60

An electrical signal other than the intended P or R wave is detected



Something other than a P or R wave was detected — rate between the two sensed beats is less than the programmed rate. Rate dropped less than paced rate.

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# TROUBLESHOOTING
## RESOLVING SENSING ISSUES: POTENTIAL CAUSES



### Heart or patient related

Inadequate cardiac signal

### Patient cable or lead related

- Loose connection at the connector block
- Heart wire (lead) dislodgement
- Inappropriate lead placement
- Insulation break or wire fracture

### Temporary pacemaker related

- Inappropriate sensitivity setting
- Inappropriate pacing mode selection
- Small sensing window
- Electrical noise

# TROUBLESHOOTING
## RESOLVING A SENSING ISSUE

- **Potential solutions for loss of sensing:**
  - Verify the cable and lead connections are secure.
  - Verify that the apparent loss of sensing is not due to blanking periods.*
  - Perform a sensing threshold test† for the affected chamber. Provide at least a 2:1 safety margin. **Note:** If the patient does not have adequate intrinsic rhythm, consult a physician before the test.
- Increase SENSITIVITY for the appropriate chamber. Slowly turn the SENSITIVITY dial counterclockwise (decrease mV until the ECG shows intrinsic activity).
- **Note:** Use caution when adjusting SENSITIVITY for patients with a history of sustained ventricular tachycardia.

* Blanking: 200 ms +5/-30 ms — after pace; 120 ms +2/-30 ms — after sense
† Instruction on sensitivity test available in section 6.3.10 Sensitivity test of the Technical Manual and video on Medtronic Academy.

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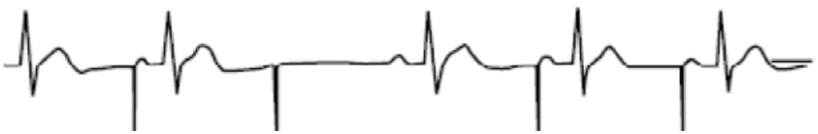
# TROUBLESHOOTING
## LOSS OF CAPTURE



Loss of atrial capture in AAI mode

Loss of ventricular capture in VVI mode

# TROUBLESHOOTING
## RESOLVING LOSS OF CAPTURE: POTENTIAL CAUSES



### Heart or patient related

- Increased stimulation threshold
- Exit block
- Tissue changes
- General metabolic imbalances
- Drug effects

### Patient cable or lead related

- Loose connection at the connector block
- Heart wire (lead) dislodgement
- Inappropriate lead placement
- Insulation break or wire fracture

### Temporary pacemaker related

Inadequate output setting (mA)

# TROUBLESHOOTING
## RESOLVING LOSS OF CAPTURE



**Possible solutions for loss of capture —** Perform one or more of the following actions to troubleshoot the cause of loss of capture:

- Check the cable connections for loose wires.
- Replace the cable if there is an insulation break or wire fracture.
- Verify that the correct chamber is being paced.
- Verify the pacing mode.
- Perform the stimulation threshold test for the affected chamber. Provide at least a 2:1 safety margin.
- If necessary, increase the output for the appropriate chamber. Slowly turn the **OUTPUT** dial clockwise until the ECG shows consistent capture.

# TROUBLESHOOTING
## KNOWLEDGE CHECK

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Patient A

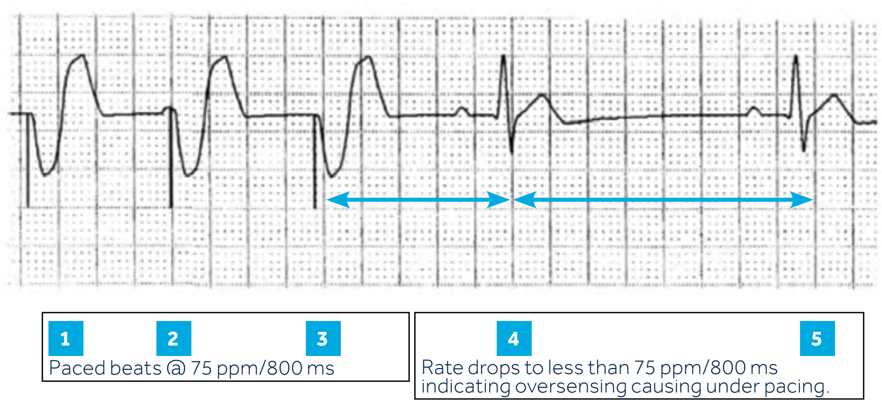
**What's happening?**

A. Undersensing with over pacing

B. Oversensing with under pacing

C. Loss of Capture

# TROUBLESHOOTING
## KNOWLEDGE CHECK




Patient A

### Patient A Feedback

- Paced rate is 800 ms/75 ppm which is seen in the intervals between beats 1-2 and 2-3.
- Interval between beats 3-4 and 4-5 is lower than the rate of 75.
- Oversensing occurred causing the rate to be lower.
- Review section: *Troubleshooting: oversensing-under pacing* for more information.

### What's happening?

A. Undersensing with over pacing

**B. Oversensing with under pacing**

C. Loss of Capture

# TROUBLESHOOTING KNOWLEDGE CHECK



**What are the next steps for Patient A?**

A. Perform a stimulation threshold test as soon as possible.

B. Verify that all lead and cable connections are secure, then consider a sensing test, observing all cautions.

C. Verify that all lead and cable connections are secure, then perform a stimulation threshold test.

D. Leave the device with current programmed settings.

# TROUBLESHOOTING KNOWLEDGE CHECK



### What are the next steps for Patient A?

A. Perform a stimulation threshold test as soon as possible.

**B. Verify that all lead and cable connections are secure, then consider a sensing test, observing all cautions.**

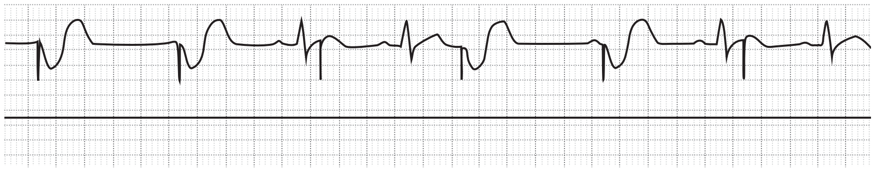
C. Verify that all lead and cable connections are secure, then perform a stimulation threshold test.

D. Leave the device with current programmed settings.

### Feedback

Please see the section: *Troubleshooting: Resolving a Sensing Issue* for more information.

# TROUBLESHOOTING
## KNOWLEDGE CHECK



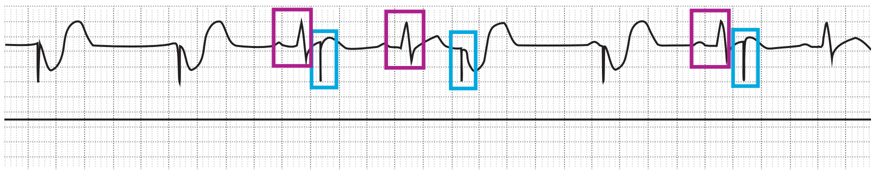
Patient B

**What's happening?**

A. Undersensing with over pacing

B. Oversensing with under pacing

C. Loss of Capture



Patient B

**Patient B Feedback**

The intrinsic beats were undersensed causing the pacing spike to occur too early.
See section: *Troubleshooting: Undersensing and Over Pacing* for more information.

☐ Undersensed beats

☐ Over paced beats due to undersensing

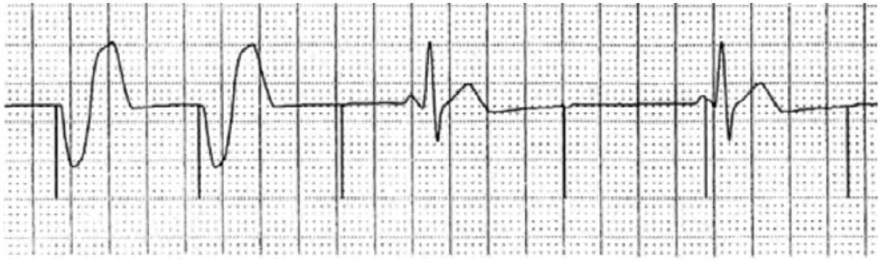
**What's happening?**

**A. Undersensing with over pacing**

B. Oversensing with under pacing

C. Loss of Capture

# TROUBLESHOOTING
## KNOWLEDGE CHECK



Patient C

### What's happening?

A. Undersensing with over pacing

B. Oversensing with under pacing

C. Loss of Capture

# TROUBLESHOOTING
KNOWLEDGE CHECK

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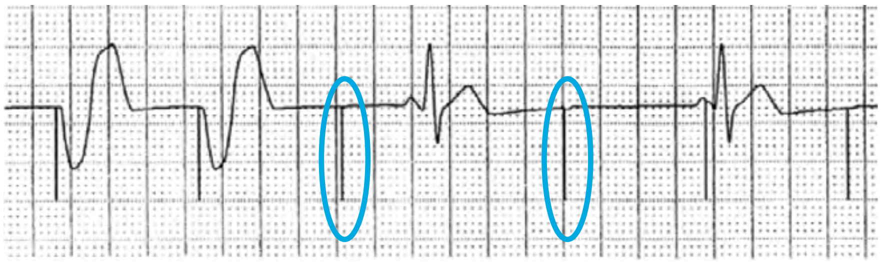

Patient C

**What's happening?**

A. Undersensing with over pacing

B. Oversensing with under pacing

**C. Loss of Capture**

## Patient C Feedback

Pacing spikes are occurring without depolarization of the cardiac tissue consistent with loss of capture.
See section:
*Troubleshooting: Loss of Capture* for more information.

# TROUBLESHOOTING KNOWLEDGE CHECK

**What are the next steps for Patient C?**

A. Verify the battery status is not blinking.

B. Verify lead and cable connections are secure.

C. Consider a threshold test.

D. Verify settings on temporary pacemaker.

E. Check patient condition.

F. All of the above.

# TROUBLESHOOTING
## KNOWLEDGE CHECK



**What are the next steps for Patient C?**

A. Verify the battery status is not blinking.

B. Verify lead and cable connections are secure.

C. Consider a threshold test.

D. Verify settings on temporary pacemaker.

E. Check patient condition.

**F. All of the above.**

**Feedback**

All of these should be considered for troubleshooting a patient with loss of capture.
Please review the section *Troubleshooting: Resolving Loss of Capture* for more information.

# RAPID ATRIAL PACING (RAP)*


OVERVIEW

BASIC OPERATION

TROUBLESHOOTING PROCESS

RAPID ATRIAL PACING (RAP)

COMPONENTS: DEVICE, BATTERY, AND CABLE

MAINTENANCE AND CLEANING

SUMMARY



RAP can be used to interrupt some types of atrial tachycardias or to induce an atrial tachycardia.

**Caution: RAP is for atrial use only.** Before enabling RAP, be sure that the atrial leads are connected to the atrium, not the ventricle.

**RAP Rates:** 80-800 paces per minute (ppm)

**Caution:**

- RAP may result in tachycardia, acceleration of existing tachycardia, or fibrillation.
- Apply high rates under careful patient monitoring and control.
- Monitor the patient's ECG and blood pressure, and ensure that defibrillation equipment is immediately available.

*Previously referred to as Burst Pacing or Overdrive Pacing

# RAPID ATRIAL PACING (RAP)*

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**Note:** When the **HOLD TO DELIVER** button is pressed and held, the temporary pacemaker waits a maximum of two pacing cycles, and then begins pacing asynchronously in the atrium (AOO pacing mode) at the selected RAP rate. RAP delivery stops when either the **HOLD TO DELIVER** button is released, or after **two minutes have passed.**

**Only use in Atrium**

*Previously referred to as Burst Pacing or Overdrive Pacing

## To perform RAP:

1. Access the RAP controls under the RAP cover.
2. Press the **ENABLE/ DISABLE** button to enter and exit RAP Standby mode.
3. Determine RAP rate by turning the Rate knob. RAP Pacing rate will appear above programmed rate.
4. Press the **Hold to Deliver** button.

# RAPID ATRIAL PACING (RAP)
## KNOWLEDGE CHECK



**Which of the following are true regarding RAP?**

A. It is only to be used when the lead is placed in the atrium.

B. RAP pacing will be maintained until five minutes after Hold to Deliver is released.

C. RAP pacing will be maintained until the Hold to Deliver button is released or a maximum of two minutes.

D. The maximum RAP rate is 320 ppm.

E. The programmed base rate and the RAP rate must match.

# RAPID ATRIAL PACING (RAP)
## KNOWLEDGE CHECK



**Which of the following are true regarding RAP?**

A. **It is only to be used when the lead is placed in the atrium.**

B. RAP pacing will be maintained until five minutes after Hold to Deliver is released.

C. **RAP pacing will be maintained until the Hold to Deliver button is released or a maximum of two minutes.**

D. The maximum RAP rate is 320 ppm.

E. The programmed base rate and the RAP rate must match.

### Feedback

- RAP may only be performed when the lead is in the atrium so you need to verify lead placement.
- RAP is only maintained when the Hold to Deliver button is held or a maximum of two minutes.
- Please review the RAP section for more information.

# COMPONENTS
## PROTECTIVE COVERS




1 Protective cover

2 RAP cover

- Both may be removed for cleaning*
- Part Number for ordering: 5340800000 (includes both covers)

*See technical manual chapter 6 for cleaning and removal of cover instructions.

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RAPID ATRIAL PACING (RAP)

COMPONENTS: DEVICE, BATTERY, AND CABLE

MAINTENANCE AND CLEANING

SUMMARY

# BATTERIES
## OVERVIEW & CHANGING BATTERIES


**Note:** Medtronic does not recommend replacing batteries while the device is turned on or connected to a patient.*

*If during an emergency situation the batteries must be replaced while the temporary pacemaker is in use, ensure that the patient is being monitored.

- Use only recommended batteries: 2 AA Alkaline batteries.
- To replace batteries, press the battery drawer latch release buttons.
- Pacing is maintained for a minimum of **30** seconds if programmed to nominal values.

# BATTERY
## BATTERY INDICATOR




- All indicator bars visible: batteries have full power.
- Flashing red light: Begins flashing when there is 24 hours of battery life remaining. Batteries should be replaced as soon as flashing light is seen.
- Check the battery status at least twice daily.

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# COMPONENTS
## COMPATIBLE CABLES



| Reusable | Disposable |
|---|---|

### Medtronic Patient Cables

**Model 5433 Family** (Reusable)

White = Ventricle

Blue = Atrium

**Model 5846 Family** (Disposable)

Blue = Atrium

White = Ventricle

**Model 5487 Family** (Disposable)

IS-1 or +/- port connections

### Medtronic Surgical Cables

**Model 5832S Family** (Reusable)

**Model 5833S Family** (Disposable)

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# COMPONENTS
## IV POLE STAND


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COMPONENTS: DEVICE, BATTERY, AND CABLE

MAINTENANCE AND CLEANING

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- An IV pole hanger is attached to the back of the device.
- Folds flat when not in use.
- Part #5340700000

# COMPONENTS
## KNOWLEDGE CHECK

**You notice the battery indicator is flashing, which of the following should you do?**

Choose all that apply.

A. Mark the patient's chart to have the batteries replaced 48 hours later.

B. Consider replacing batteries immediately.

C. Use two AA lithium batteries when replacing batteries.

D. Use two Alkaline batteries when replacing batteries.

# COMPONENTS
## KNOWLEDGE CHECK

**You notice the battery indicator is flashing, which of the following should you do?**

Choose all that apply.

A. Mark the patient's chart to have the batteries replaced 48 hours later.

**B. Consider replacing batteries, immediately.**

C. Use two AA lithium batteries when replacing batteries.

**D. Use two Alkaline batteries when replacing batteries.**

### Feedback

- Batteries should be replaced as soon as possible when the flashing battery indicator is seen.
- Only alkaline batteries should be used. See section: *Components: Batteries: Overview and Changing Batteries and Battery Indicator.*

# COMPONENTS
## KNOWLEDGE CHECK



**Which of the following are true regarding compatible cables? Choose all that apply.**

A. Only surgical cables may be used.

B. Only reusable patient cables may be used.

C. Patient and surgical cables may be used.

D. Reusable and disposable cables may be used.

# COMPONENTS
## KNOWLEDGE CHECK



**Which of the following are true regarding compatible cables? Choose all that apply.**

A. Only surgical cables may be used.

B. Only reusable patient cables may be used.

**C. Patient and surgical cables may be used.**

**D. Reusable and disposable cables may be used.**

### Feedback

The 53401 Temporary External pacemaker is compatible with surgical and patient cables as well as reusable and disposable models.
See section:
*Components: Compatible Cables* for more information.

# MAINTENANCE AND CLEANING

## CLEANING

- Clean the temporary pacemaker before each use for a new patient.
- Return it to Medtronic for service if the temporary pacemaker has blood or soil ingress in its battery compartment, cable ports, or under the knobs.
- Do not expose the temporary pacemaker to ethers, acetone, chlorinated solvents, or disinfectants that are not included in the manual instructions.
- The Technical Manual section 6 covers cleaning information.
- For more information on cleaning and disinfecting medical devices, visit the CDC and HICPAC website: cdc.gov/hicpac/Disinfection_Sterilization/14_00ReuseMedicalDevices.html.

# MAINTENANCE AND CLEANING
## CLEANING

- Clean all surfaces of the temporary pacemaker thoroughly using a 70% isopropyl alcohol prep pad.
- Disinfecting — 15 minutes of wet/damp exposure to 70% isopropyl alcohol is needed for disinfection. To achieve 15 minutes of moistened exposure, wrap the temporary pacemaker and covers in cloths dampened with 70% isopropyl alcohol and place them in a sealed bag or container.
- Do not expose the temporary pacemaker to ethers, acetone, chlorinated solvents, or disinfectants.
- Do not immerse the temporary pacemaker in water or cleaning agents. Severe damage to the temporary pacemaker may occur.
- Do not use automated machine washers. Do not sterilize the temporary pacemaker by ethylene oxide, gamma radiation, or steam-sterilization (autoclave). Damage to the temporary pacemaker may occur using these methods.

# MAINTENANCE AND CLEANING
## MAINTENANCE

- Safety and technical checks should be carried out on the temporary pacemaker periodically, and after any malfunction or accident. Medtronic recommends carrying out safety and technical checks at least once every 12 months.
- Contact your Medtronic sales or service representative for additional service or training. If a checkout or test of a pacemaker indicates improper operation, the unit **should be returned to Medtronic for evaluation and repair.**
- For additional safety and technical information and testing refer to the Technical Manual section 6.3, page 40.

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RAPID ATRIAL PACING (RAP)

COMPONENTS: DEVICE, BATTERY, AND CABLE

MAINTENANCE AND CLEANING

SUMMARY


**What should be done if the 53401 temporary pacemaker has been contaminated with blood into the battery component, cable ports, or under the knobs?**

A. Clean by soaking in a chlorinated solution.

B. Clean by soaking in acetone.

C. Return to Medtronic for cleaning.

# MAINTENANCE AND CLEANING
## KNOWLEDGE CHECK



**What should be done if the 53401 temporary pacemaker has been contaminated with blood into the battery component, cable ports, or under the knobs?**

A. Clean by soaking in a chlorinated solution.

B. Clean by soaking in acetone.

**C. Return to Medtronic for cleaning.**

**Feedback**

The 53401 Temporary External Pacemaker should be returned to Medtronic for cleaning if it has been contaminated with blood in the battery component, cable ports, or under the knobs.
See section:
*Maintenance and Cleaning: Cleaning* for more information.

# COMPONENTS
## KNOWLEDGE CHECK



**You are notified that the 53401 Temporary Pacemaker was dropped on the floor after being used on a patient. What is the best choice of action?**

A. Clean and return to storage.

B. Ensure safety and technical checks are completed before the unit is used again.

# COMPONENTS
## KNOWLEDGE CHECK



**You are notified that the 53401 Temporary Pacemaker was dropped on the floor after being used on a patient. What is the best choice of action?**

A. Clean and return to storage.

**B. Ensure safety and technical checks are completed before the unit is used again.**

**Feedback**

Dropping on the floor is considered an accident. Safety and technical checks should be performed after the device has been dropped.
See section:
*Maintenance and Cleaning: Maintenance* for more information.

# 53401 IN-SERVICE TRAINING
## SUMMARY

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- The 53401 Temporary External Pacemaker provides constant current single chamber pacing in synchronous and asynchronous pacing.
- Device runs on two AA (LR6) alkaline batteries.
- Temporary heart wires connect via reusable or disposable patient cables.
- Sensing and capture issues may be resolved by following the troubleshooting process.
- Rapid Atrial Pacing (RAP) is available when lead is placed in the atrium.
- Protective cover, RAP cover, and IV pole hanger may be ordered separately.
- Device may be cleaned by using 70% isopropyl alcohol.
- Device should be returned to Medtronic if contaminated with blood into the battery component, cable ports, or under the knobs.

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COMPONENTS: DEVICE, BATTERY, AND CABLE

MAINTENANCE AND CLEANING

SUMMARY

**Indications, Safety, and Warnings**

**If you are located in the United States,** please refer to the brief statement(s) below to review applicable indications, safety, and warning information. See the device manual for detailed information regarding the implant procedure, indications, contraindications, warnings, precautions, and potential complications/adverse events. For further information, please call Medtronic at 1-763-514-4000 and/or consult the Medtronic website at medtronic.com.

**If you are located outside the United States**, see the device manual for detailed information regarding the implant procedure, indications, contraindications, warnings, precautions, and potential adverse events. For further information, contact your local Medtronic representative and/or consult the Medtronic website at medtronic.com.

www.medtronic.com/manuals

Consult instructions for use on this website. Manuals can be viewed using a current version of any major Internet browser. For best results, use Adobe Acrobat® Reader with the browser.

**Important Reminder:** This information is intended only for users in markets where Medtronic products and therapies are approved or available for use as indicated within the respective product manuals. Content on specific Medtronic products and therapies is not intended for users in markets that do not have authorization for use.

**Brief Statement**
**Model 53401 Temporary External Pacemaker**
**Intended Use:** The Medtronic Model 53401 Temporary External Pacemaker is intended to be used in conjunction with a cardiac pacing lead system for temporary atrial or ventricular pacing in a clinical environment by trained personnel. The temporary pacemaker can be used where short-term demand (synchronous) or asynchronous pacing is indicated for therapeutic, prophylactic or diagnostic purposes. The temporary pacemaker must be used in an environment where the patient is monitored continuously to ensure that it is operating properly and delivering appropriate therapy to the patient. **Contraindications:** There are no known contraindications to the use of temporary pacing as a means to control the heart rate. The patient's age and medical condition, however, may dictate the type of temporary pacemaker and lead system used by the physician. Single chamber atrial pacing is contraindicated in the presence of AV conduction disorders. Asynchronous pacing is contraindicated in the presence of intrinsic cardiac rhythms. Atrial high-rate burst pacing therapy is intended for use in the atrium only. High-rate burst pacing in the ventricle may result in life-threatening arrhythmias. The temporary pacemaker is MR Unsafe. **Warnings/Precautions:** Monitor the patient continuously while the temporary pacemaker is in use to ensure it is operating properly and delivering appropriate therapy to the patient. ECG monitoring should be in use and defibrillating equipment should be placed on standby and be kept immediately available during pacing lead insertion, pulse generator connection and adjustment, measurements of stimulation thresholds or sensed potentials, and application of antitachycardia burst therapy. Use of high rates in the atrium may result in accidental conduction to the ventricle. Defibrillation equipment should be kept immediately available during high-rate pacing. Operational failure of the temporary pacemaker can occur as the result of battery depletion, mishandling, or random component failure. Complications related to the use of temporary external pacemakers include, but are not limited to asystole following abrupt cessation of pacing, inhibition, and reversion. Potential complications related to the use of pacing lead systems with the temporary pacemaker include, but are not limited to myocardial irritability resulting in fibrillation, infarction, pericarditis, rejection, muscle and never stimulation, and infection. Complications may result due to inhibition or reversion of the pacemaker in the presence of strong electromagnetic interference. Whenever possible, for the safety of the patient, disconnect the temporary pacemaker from the implanted lead system before defibrillating or cardioverting. Excessive defibrillation energy can damage the temporary pacemaker. This can result in a large current flowing through the implanted lead system and temporary pacemaker, which could reduce intended defibrillation energy delivered to the patient or cause myocardial damage. A lead with extension cable constitutes a direct, low-resistance current path to the myocardium. During connection and testing procedures, only batterypowered instrumentation should be used. Extreme caution must be taken to properly ground all line-powered equipment used in the vicinity of the patient. Electrosurgical units can cause tachyarrhythmias by inducing current on the leads. Improper connection, displacement or fracture of leads or cables may result in pacemaker system failure. Inspect leads and cables for damage before each use. The pacing lead system may cease to function at any time due to improper connections or lead-related problems such as displacement or fracture. Do not modify the temporary pacemaker. Modifications could impact the temporary pacemaker effectiveness and adversely affect patient safety.

*See the device manual for detailed information regarding the instructions for use, indications, contraindications, warnings, precautions, and potential complications/ adverse events. For further information, please call Medtronic at 1-800-328-2518 and/or consult the Medtronic website at* medtronic.com.

**Caution:** Federal law (USA) restricts this device to sale by or on the order of a physician.

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